

LEMON-Factor: Interpretable Factor-Level Semantic Coverage for Graph-to-Text Evaluation

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Abstract. Graph-to-text systems should preserve the meaning of each input relation, not merely mention the correct entities or produce fluent text. Existing automatic metrics for RDF-to-text and data-to-text generation are mostly reference-based and holistic, while recent faithfulness metrics often operate at the level of whole triples. We propose LEMON-FACTOR, a diagnostic evaluation framework that decomposes source predicates into weighted semantic factors and checks how these factors are reflected in a verbalization. The method is source-directed: it starts from graph predicates and their argument roles, not from text-only atomic claims. We implement a reproducible pilot on a stratified WebNLG subset, construct a predicate inventory, bootstrap a controlled factor schema, generate and adjudicate predicate decompositions, and compute deterministic graph-text coverage. In the first coverage run, 140 of 308 development edges were scored and LEMON-FACTOR coverage was 0.368. After error-driven decomposition and lexical-cue expansion, all 308 edges were scored and coverage rose to 0.791, while exact entity-label overlap remained unchanged at 0.921. The results show that factor-level coverage exposes a diagnostic layer that surface overlap does not capture. The current study is a pilot: synthetic LLM adjudication and rule-based expansion are not treated as human gold.

Keywords: Graph-to-text evaluation · Semantic coverage · Knowledge graphs · WebNLG · Predicate decomposition · LLM adjudication

1 Introduction

Knowledge graphs encode facts as typed relations between entities. In RDF-style graph-to-text generation, a system is given triples such as (s, p, o) and must produce a natural-language verbalization that is fluent and faithful to the source graph. The central evaluation problem is therefore not only whether the generated sentence contains the right entity names, but whether it preserves the meaning of each predicate and the roles implied by that predicate.

This distinction matters because predicate meaning is rarely monolithic. A relation such as *birthPlace* invokes at least a biographical event, a person-like subject, and a location-like object. A relation such as *leaderTitle* invokes political authority, an office or title, and often a governed place or institution. A generated

text may mention both entities while omitting the relation, may express the relation through paraphrase, or may verbalize only one role of the predicate. Holistic metrics collapse these cases into a single sentence-level score.

The WebNLG line of shared tasks established a widely used benchmark for mapping sets of RDF triples to text [1, 2]. Its evaluation practice combines automatic generation metrics, such as BLEU, METEOR, chrF++, TER, BERTScore, and BLEURT, with human criteria such as data coverage, relevance, correctness, and fluency. These metrics and criteria are valuable, but they do not provide a within-predicate audit trail: they do not say which semantic component of which source predicate was preserved, omitted, or only weakly expressed.

We propose LEMON-FACTOR, an interpretable, source-directed coverage framework. Instead of decomposing generated text into claims, LEMON-FACTOR decomposes graph predicates into weighted semantic factors. Each factor has a role, such as *predicate meaning*, *subject domain*, or *object domain*, and a salience weight. Coverage is computed by checking evidence for each factor in the generated text. The result is both a score and a diagnostic explanation at the edge and factor level.

This paper reports a reproducible pilot on a stratified WebNLG subset. The pilot has four contributions. First, it defines a controlled factor schema for RDF predicates grounded in predicate-role decomposition. Second, it proposes a weighted factor-level coverage score for graph-text evaluation. Third, it implements a deterministic coverage baseline with explicit lexical evidence and audit trails. Fourth, it reports an error-driven expansion experiment showing that missing predicate decompositions, not just text quality, strongly affect factor-level coverage.

The scope is deliberately limited. Predicate decompositions are initially bootstrapped by rules and synthetic LLM adjudication rather than by human domain experts. The coverage scorer is lexical and deterministic rather than neural. These design choices make the first pilot reproducible and inspectable, but they do not constitute human validation.

2 Related Work

2.1 RDF-to-text benchmarks and standard metrics

The WebNLG Challenge introduced RDF-to-text generation as a benchmark task: systems receive DBpedia-derived RDF triples and generate fluent verbalizations [1]. Later WebNLG+ tasks extended the setting to bilingual and bidirectional generation [2]. The standard automatic evaluation stack includes n-gram and edit-distance metrics as well as neural reference metrics. BLEU remains a conventional reference point [3], while BERTScore [4] and BLEURT [5] add contextual or learned similarity. These metrics compare generated text with reference text and therefore remain mostly holistic.

2.2 Data coverage and faithfulness metrics

A second line of work evaluates generated text against the source data. Wiseman et al. proposed information-extraction-based evaluation for data-to-document generation [6]. PARENT aligns entailed n-grams with semi-structured source tables and references [7]. Data-QuestEval uses question generation and answering for reference-less data-to-text evaluation [9]. AlignScore learns a unified alignment function for factual consistency [10]. The nearest graph-to-text predecessor is FactSpotter, which decides whether a given triple is present in the generated text and reports strong correlation with human annotations for correctness, coverage, and relevance [8]. LEMON-FACTOR complements FactSpotter by moving from binary triple-level decisions to weighted factor-level decisions inside a predicate.

2.3 Atomic fact and semantic graph evaluation

Atomic factuality metrics such as FActScore decompose generated text into atomic claims and verify each claim against a knowledge source [11]. VERISCORE extends this line to verifiable claims in long-form generation [12]. These methods are text-directed: the decomposition target is the generated text. LEMON-FACTOR is source-directed: the decomposition target is the graph predicate.

Semantic graph evaluation metrics such as Smatch compare semantic graphs, especially AMR graphs, by maximizing triple overlap under variable alignment [13]. S²match and Smatch++ add softer and more standardized variants of graph matching [14, 15]. These metrics are important for parser evaluation, but they require a semantic graph representation of the generated text. LEMON-FACTOR avoids inverse parsing in the first baseline and instead checks factor evidence directly in the text.

2.4 Predicate decomposition and synthetic adjudication

The theoretical motivation for factor decomposition comes from predicate-argument and frame-based semantics. FrameNet represents lexical meanings through frames and frame elements [16]; PropBank annotates predicate-argument structures over corpora [17]; VerbNet provides broad-coverage verb classes with semantic roles [18]. These resources support the view that predicate meaning can be represented through typed roles rather than through a single label.

The present pilot uses LLM-generated and LLM-adjudicated predicate decompositions as a bootstrap. This is not equivalent to expert annotation. Prior work on LLM-as-a-judge documents useful agreement with human preferences as well as position, verbosity, and self-enhancement biases [20]. Surveys and empirical studies argue that LLM judges must be calibrated against human judgments before being treated as primary evaluators [21, 22]. We therefore mark our adjudicated resource as synthetic.

Table 1. Positioning of LEMON-FACTOR relative to related evaluation lines.

Line of work	Main object measured	Limitation addressed by LEMON-FACTOR
Reference metrics	Similarity to reference text	No source-predicate audit trail
Source faithfulness	Source-text consistency	Usually triple- or sentence-level, not factor-level
Atomic fact metrics	Text-derived claims	Decompose generated text, not source predicates
Semantic graph metrics	Graph-graph overlap	Require parsing text into a graph
Semantic role sources	Predicate roles and frames	Resources, not graph-text coverage metrics

3 Method

3.1 Graph-text representation

Each example is represented as a graph-text pair. The graph contains directed edges $e = (s, p, o)$, where s and o are entity labels and p is a predicate. The text is the reference or system verbalization to be evaluated. The current pilot uses WebNLG references as the verbalizations, but the scorer can be applied to any graph-text pair in the same normalized format.

3.2 Semantic factor schema

A predicate decomposition maps each predicate p to a list of weighted components:

$$D(p) = \{(f_i, r_i, w_i)\}_{i=1}^n,$$

where f_i is a semantic factor, r_i is its role, and $w_i > 0$ is its salience weight. Roles include *predicate meaning*, *subject domain*, *object domain*, *value domain*, and auxiliary background or modifier roles. The seed schema contains general factors such as *entity*, *person*, *place*, *organization*, *time*, *quantity*, and relation factors such as *biographical relation*, *location relation*, *part-whole relation*, *political relation*, and *creative relation*.

The factorization is role-aware. For *birthPlace*, the predicate meaning is biographical, the subject domain is person-like, and the object domain is place-like. For *orbitalPeriod*, the predicate meaning is astronomical and the object domain is temporal or quantitative. This representation permits partial credit: a text may mention the subject and object while failing to verbalize the relation.

3.3 Factor evidence and coverage score

For a text t , an edge e , and a component (f_i, r_i, w_i) , the evidence function $E(f_i, r_i, e, t) \in [0, 1]$ checks whether the text contains evidence for the component. In the current deterministic baseline, subject-domain and object-domain

Table 2. Pilot WebNLG statistics used in the current draft.

Split	Examples	Edges	Avg. edges	Predicates	Categories
Train	200	568	2.84	162	16
Development	100	308	3.08	119	16

components are supported by normalized label matches or token overlap. Predicate-meaning components are supported by lexical cues associated with the predicate or factor, such as *born in* for *birthPlace* or *part of* for *isPartOf*.

The edge-level score is a weighted mean:

$$C(e, t) = \frac{\sum_i w_i E(f_i, r_i, e, t)}{\sum_i w_i}.$$

Example-level coverage is the mean of edge-level scores over the graph. Corpus-level coverage is the mean over examples or edges, accompanied by missing-decomposition rate and low-coverage edge count.

3.4 Decomposition resources

The pipeline constructs decomposition resources in stages. First, WebNLG predicates are converted to candidate lexical factors by splitting camelCase and slash-separated names. Second, seed rules map frequent predicates to controlled factors. Third, optional LLM calls through a fixed JSON schema generate candidate decompositions. Fourth, a stronger LLM adjudicator produces a synthetic adjudicated reference. Finally, error-driven expansion adds decompositions for missing development predicates and extends lexical cues. All synthetic and expanded resources are marked as non-gold.

4 Data and Experimental Resources

4.1 WebNLG pilot

The experiments use a stratified English WebNLG pilot. The train split contains 200 examples and the development split contains 100 examples. Both splits cover 16 WebNLG categories. The development set contains 308 graph edges. The pilot is intentionally small: it is designed to validate the scoring pipeline and to make error analysis inspectable before scaling to larger WebNLG subsets or domain-specific graphs.

4.2 Synthetic predicate decompositions

A seed factor inventory was built from train predicates and categories. LLM-assisted candidate generation was then tested with two models. The LLM stage is model-agnostic: all models receive the same factor schema, prompt template,

Table 3. LLM decomposition candidate quality against the seed reference. This is not human validation.

Model	Items	Factor P	Factor R	Factor F1	Role Acc.
Claude 3.5 Haiku	19	0.754	0.807	0.775	0.596
Llama 3.1 70B Instruct	19	0.798	0.798	0.795	0.614

Table 4. Synthetic adjudication diagnostics.

Metric	Value
Items	29
LLM candidate coverage	0.621
Predicates with LLM candidates	18
Predicates without LLM candidates	11
Mean confidence	0.830
Modified decisions	19
Accepted seed decisions	10

and JSON output constraints. In the sanity run, the two models achieved similar factor F1 against the rule-based seed reference, while role accuracy remained lower, indicating that role assignment is the main unstable part of the task.

A synthetic adjudicator was then applied to 29 predicates. The synthetic reference had LLM candidate coverage of 0.621: 18 predicates had LLM candidates and 11 predicates did not. The mean adjudicator confidence was 0.830. This resource is useful for development, but it is not a substitute for human expert annotation.

5 Experiments

5.1 Experiment 1: initial factor coverage

The first experiment scores the WebNLG development set using the synthetic adjudicated predicate decomposition dictionary and the seed lexical cue dictionary. We report five quantities: exact entity-label coverage, predicate lexical cue coverage, weighted LEMON-FACTOR coverage, missing decomposition rate, and the number of low-coverage edges under a threshold of 0.5.

5.2 Experiment 2: error-driven expansion

The initial run exposes predicates without decompositions. We therefore run a missing-predicate analysis, generate rule-based or fallback decompositions for the missing predicates, extend the lexical cue dictionary, and recompute coverage. This experiment tests whether LEMON-FACTOR can be improved through interpretable error-driven resource expansion.

5.3 Experiment 3: lexical baseline comparison

The third experiment compares LEMON-FACTOR to lightweight lexical controls. Exact entity-label overlap checks whether entity labels are mentioned.

Table 5. Coverage before and after error-driven expansion.

Metric	Initial	Expanded	Delta
Scored edges	140	308	+168
Missing decomposition rate	0.5455	0.0000	-0.5455
Predicate cue coverage	0.3149	0.6526	+0.3377
LEMON-FACTOR coverage	0.3683	0.7915	+0.4232
Low-coverage edges	191	41	-150

Predicate cue coverage checks explicit relation cues. Graph-text token Jaccard and graph-text token cosine compare a bag-of-words graph lexicalization with the text. These controls are not intended as strong semantic metrics; they make explicit whether LEMON-FACTOR adds a diagnostic dimension beyond surface overlap.

6 Results

6.1 Initial and expanded coverage

The initial run scored 140 of 308 development edges. The missing decomposition rate was 0.5455, exact label coverage was 0.9215, predicate cue coverage was 0.3149, and LEMON-FACTOR coverage was 0.3683. This confirms that entity labels are often preserved in WebNLG references, while predicate-level semantic coverage cannot be established for many edges with the initial resource.

After error-driven expansion, all 308 edges were scored. Missing decomposition rate fell to 0.0. Predicate cue coverage increased to 0.6526 and LEMON-FACTOR coverage increased to 0.7915. The low-coverage edge count dropped from 191 to 41.

The increase should not be interpreted as proof that generated texts are universally faithful. It shows that the first score was strongly constrained by decomposition and cue coverage. Once all predicates receive decompositions and stronger cues, the scorer can measure relation-level evidence more completely.

6.2 Comparison with lexical controls

Table 6 summarizes the baseline comparison. Exact entity-label overlap is high and unchanged by expansion. Graph-text token cosine and Jaccard are also unchanged, because they do not use predicate decompositions. Predicate cue coverage and LEMON-FACTOR coverage both increase after expansion, but LEMON-FACTOR remains distinct because it weights role-aware factors, not only lexical cues.

The main observation is that surface/entity overlap already appears high and does not explain the change in LEMON-FACTOR. The proposed metric is therefore sensitive to the completeness of predicate-level semantic decompositions and relation evidence, not only to whether entity names are repeated.

Table 6. Baseline comparison on the WebNLG development pilot.

Method	Initial	Expanded	Delta
Exact entity-label overlap	0.9215	0.9215	0.0000
Predicate lexical cue coverage	0.3149	0.6526	+0.3377
Graph-text token cosine	0.7357	0.7357	0.0000
Graph-text token Jaccard	0.6233	0.6233	0.0000
LEMONT-FACTOR coverage	0.3683	0.7915	+0.4232

7 Error Analysis

The first run produced a large number of low-coverage edges primarily because the decomposition dictionary did not cover many development predicates. This is a useful failure mode: it identifies precisely which predicates require analysis before the metric can be trusted. The expansion step turned this diagnostic signal into new decomposition and lexical-cue resources.

After expansion, the remaining low-coverage cases fall into four broad classes. First, some edges have weak lexical evidence because the relation is paraphrased without one of the current cues. Second, some fallback decompositions are too generic: they allow scoring but do not yet represent domain-specific roles well. Third, some texts mention entities but leave the predicate relation implicit. Fourth, synthetic adjudication may have introduced noisy factor choices for predicates that require expert domain knowledge.

This behavior is central to the method. A low score is not only a scalar failure; it can be traced to a predicate, factor, role, and evidence type. This makes the metric suitable for dataset diagnostics, generation-model error analysis, and iterative improvement of semantic resources.

8 Limitations

The most important limitation is the absence of human expert validation. The current factor decompositions are produced by deterministic rules, LLM candidates, and a synthetic LLM adjudicator. LLM-as-a-judge research documents systematic biases, so these decompositions must not be described as expert gold. The current resource should be treated as a provisional bootstrap.

The second limitation is lexical evidence. The deterministic scorer can find explicit labels and known cues, but it can miss paraphrases, implicit relations, and discourse-level entailment. A future version should add embedding or entailment evidence while keeping the same factor-level audit trail.

The third limitation is scale. The pilot covers 100 development examples and 308 edges. Larger WebNLG experiments and domain-specific datasets are needed. The next validation step should measure correlation between LEMON-FACTOR and human data-coverage judgments, and compare directly with FactSpotter, Data-QuestEval, and neural reference metrics.

Finally, expansion can inflate scores if fallback decompositions are too permissive. For this reason we report initial and expanded runs separately and mark

expanded results as a deterministic baseline rather than as human-validated truth.

9 Conclusion

We introduced LEMON-FACTOR, a source-directed factor-level coverage framework for graph-text evaluation. The method decomposes RDF predicates into weighted semantic factors and scores whether each factor is reflected in the text. In a WebNLG pilot, LEMON-FACTOR exposed the difference between high entity-label overlap and lower predicate-level semantic coverage. Error-driven expansion reduced missing decomposition rate from 0.5455 to 0.0 and raised LEMON-FACTOR coverage from 0.3683 to 0.7915, while surface lexical baselines stayed unchanged.

The result supports the main claim of the pilot: factor-level semantic coverage provides interpretable diagnostic information that is not available from holistic reference metrics or simple lexical overlap. Future work will replace synthetic adjudication with human expert validation, add paraphrase-sensitive evidence, and compare the metric against human data-coverage judgments at larger scale.

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